Ahmedabad University

S2 GROUP- 8

NETWORKING

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PROJECT CONTEXT AND MOTIVATION

- **Title:** Probabilistic Task Offloading in D2D Edge Computing.
- **Domain:** Device-to-Device Edge Computing Networks.
- **Real-World context:** Variable task execution and network delays.
- **Why uncertainty exists:** Task size, device load, and network conditions vary dynamically.

PROJECT OBJECTIVE AND SCOPE

- **Objective:** Maximize Task Completion Rate within deadlines.
- **Decision problem:** Where to offload computation tasks.
- **What is predicted:** Probability of successful task completion.
- **Scope:** Probabilistic, game-based offloading decisions

PROJECT SYSTEM OVERVIEW

- Task generation at user devices
- Probability estimation of task completion
- Game-theoretic offloading decision
- Task execution (local / remote device)
- Deadline success evaluation

SOURCES OF UNCERTAINITY IN THE PROJECT

- Variable task processing time
- Queueing and contention delays
- Device heterogeneity
- Fluctuating transmission delay

KEY RANDOM VARIABLES AND THEIR ROLES

- **Task Processing Time (T):** Execution duration
- **Task Arrival Rate (λ):** Random task generation
- **Task Completion Time:** Total delay (queue + execution + transmission)
- **Completion Indicator:** Deadline met or not (0/1)
- **Device Workload:** Dynamic system load

PROBABLISTIC MODELS AND ASSUMPTIONS

- Task processing modeled using **Exponential distribution**
- Task arrivals modeled as **Poisson process**
- **Independent arrivals** assumption
- **i.i.d. service times** for same task types
- **Memoryless property** for processing time
- Decisions converge to **Nash Equilibrium**

PROBABILISTIC REASONING AND INFERENCE LOGIC

Uncertainty Propagation

- Task processing time is uncertain
- Queueing and transmission delays vary
- Completion time and deadline satisfaction are probabilistic

Probabilistic Decision Logic

- Devices estimate probability of meeting deadlines
- Offloading maximizes completion probability
- Conditional probabilities avoid congestion

Reasoning Pathway

- Random task arrival → probabilistic processing → queueing delay
- Completion probability computed (CDF)
- Best offloading target selected
- Iterative updates converge to Nash Equilibrium

OPERATIONALIZING PROBABILSTIC REASONING IN THE PROJECT

From Theory to Practice

- Task times modeled from measured data
- Completion probabilities computed via queueing CDFs
- Offloading decisions maximize deadline success probability

Evaluation Metrics

- Task Completion Rate = probability of meeting deadline
- Overall performance = aggregated completion probabilities
- Convergence indicates stable (Nash) behavior

Simplifying Assumptions

- Poisson task arrivals, independent workloads
- Known application types
- M/G/1 queueing model and short-term stability

CURRENT LIMIITATIONS AND CONCEPTUAL GAPS

Model Limitations

- Single-server (M/G/1) abstraction
- Heavy reliance on accurate TPT fitting

Questionable Assumptions

- Poisson task arrivals
- Known application types
- Deterministic transmission delays

Conceptual Gaps

- Nash Equilibrium ≠ global optimality
- Limited analysis of overhead vs accuracy trade-offs

PLANNED REFINIMENTS AND ROLE COORDINATION

Next Milestone

- Refine probabilistic models with more data
- Improve realism of arrival and transmission models

Tasks Ahead

- Relax Poisson and single-server assumptions
- Explore adaptive or learned TPT distributions

Role Coordination

- Scribe – Falak Shah, Nikunj Kalariya; Video/PPT – Ishani Prajapati, Urvil Chodvadiya; Project Manager (Concept Map & GitHub) – Manthan Pancholi.

SUMMARY OF GROUP LEVEL UNDERSTANDING

- Task processing times are uncertain
- Probabilistic models handle this variability
- Queueing links uncertainty to completion probability
- Offloading maximizes deadline success probability
- Device decisions are interdependent
- Game theory ensures stable operation